Day 2 Activity: School Scheduling & Classroom Protocols (Intro Stats)

Name: _____ Date: _____

Scenario: You are the coordinator for a K–12 school. Your job involves building schedules, planning student support pathways, and managing classroom procedures. Use counting principles to answer each question.

Part 1: Fundamental Counting Principle (FCP)

Show your setup as a product (e.g., choices × choices × choices) and give the final number.

1) Teacher Assignment Assign 1 math teacher from 4, 1 ELA teacher from 6, and 1 counselor from 3 for an after-school program. How many different staffing combinations are possible? Work:

2) Student Support Pathway A student can meet with 1 of 3 interventionists, then attend 1 of 2 study labs, then check in with 1 of 4 peer tutors. How many possible support pathways are there? Work:

3) Classroom Procedure A teacher can start class with 1 of 2 warm-ups, then use 1 of 3 discussion protocols, and close with 1 of 2 exit ticket formats. How many procedure sequences exist? Work:

Part 2: Simplifying Factorial Expressions (Numbers Only)

These mirror the calculations inside permutation and combination formulas. Simplify each exactly (cancel before multiplying when possible).

4) $8!/5! =$

5) $10!/(7! \cdot 3!) =$

6) $12!/9! =$

7) $9!/(6! \cdot 3!) =$

8) $15!/(12! \cdot 3!) =$

Part 3: Permutations (Order Matters)

Identify the correct model and compute.

9) Presentation Order Five student groups must present in a specific sequence during class. In how many different orders can the 5 groups present? Model/Work:

10) Assembly Roles From a team of 8 student leaders, assign 3 distinct roles: Emcee, Tech Lead, and Stage Manager. In how many ways can these roles be filled? Model/Work:

Part 4: Combinations (Order Does Not Matter)

Identify the correct model and compute.

11) Committee Selection Select 4 teachers from a group of 10 to serve on the School Climate Committee. How many different committees are possible? Model/Work:

12) Field Trip Chaperones Choose 5 parent chaperones from 12 volunteers for a field trip. How many different groups of 5 can be selected? Model/Work:

Optional Reflection (If time remains) • When should you use FCP instead of permutations or combinations? • How do you recognize when order matters in real school contexts (roles vs. groups)?